

Practical 13: Comparative Analysis of Forecasting Models

Question: Using the same dataset, fit the following forecasting models: (a) Simple Exponential Smoothing, (b) Holt's Method, (c) Holt-Winters Method, (d) ARIMA. Calculate MAPE or RMSE for each method and identify which model performs best with justification.

13.1 Objective

- To implement multiple forecasting techniques (SES, Holt's, Holt-Winters, ARIMA) on a single dataset.
- To evaluate the performance of each model using Root Mean Squared Error (RMSE).
- To determine the best-fitting model for the data.

13.2 Theory

- **SES:** Suitable for data with no trend or seasonality (Level only).
- **Holt's Linear:** Handles data with Trend but no seasonality (Level + Trend).
- **Holt-Winters:** Handles data with both Trend and Seasonality (Level + Trend + Seasonality).
- **ARIMA:** Models correlations in the data (AutoRegressive + Moving Average).
- **RMSE (Root Mean Squared Error):** Measures the average deviation of the forecasts from the actual values. Lower is better.

13.3 Code Implementation

```
ses_model = SimpleExpSmoothing(train).fit()
pred_ses = ses_model.forecast(12)

# (b) Holt's Linear (Trend)
holt_model = ExponentialSmoothing(train, trend='add', seasonal=None).fit()
pred_holt = holt_model.forecast(12)

# (c) Holt-Winters (Trend + Seasonality)
hw_model = ExponentialSmoothing(train, trend='add', seasonal='mul',
seasonal_periods=12).fit()
pred_hw = hw_model.forecast(12)

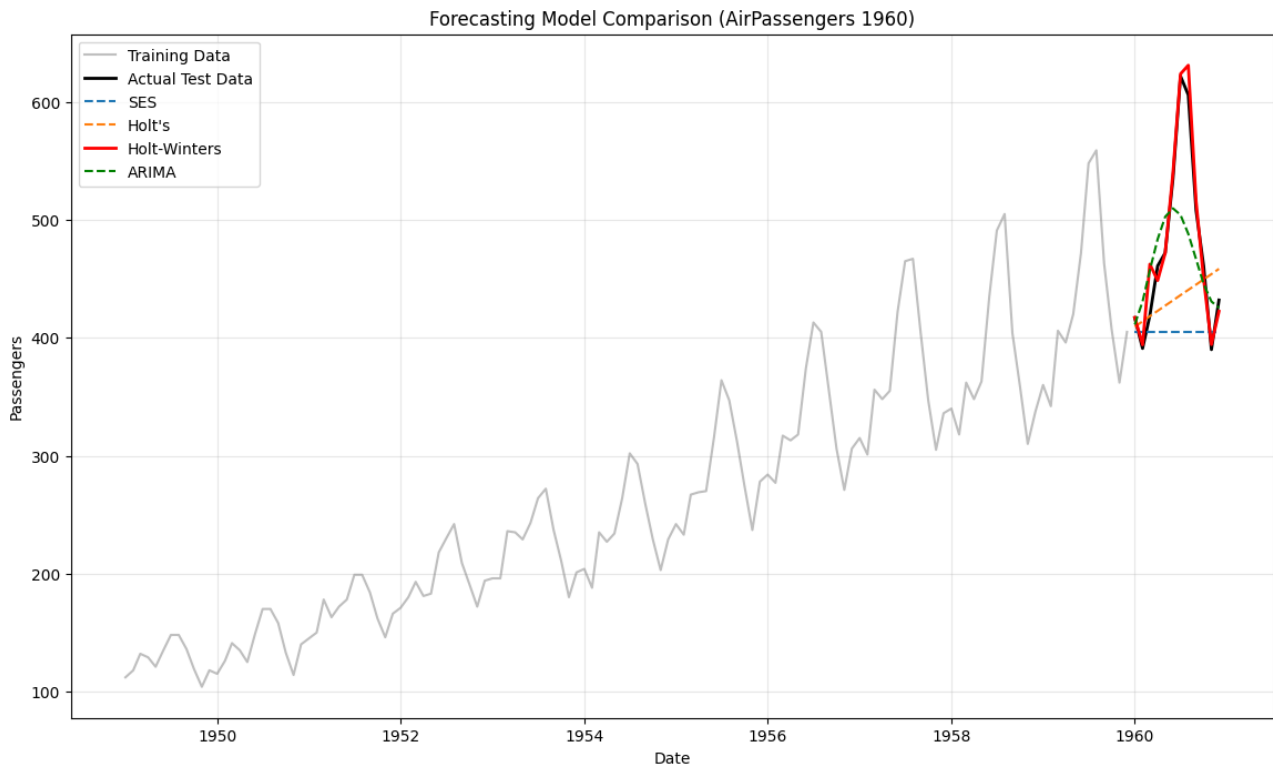
# (d) ARIMA (Example Order 1,1,1)
arma_model = ARIMA(train, order=(1,1,1)).fit()
pred_arma = arma_model.forecast(12)

# 3. Calculate RMSE
rmse_ses = np.sqrt(mean_squared_error(test, pred_ses))
rmse_holt = np.sqrt(mean_squared_error(test, pred_holt))
rmse_hw = np.sqrt(mean_squared_error(test, pred_hw))
rmse_arma = np.sqrt(mean_squared_error(test, pred_arma))

# 4. Compare Results
results = pd.DataFrame({
    'Model': ['SES', 'Holt', 'Holt-Winters', 'ARIMA'],
```

```
'RMSE': [rmse_ses, rmse_holt, rmse_hw, rmse_arima]
})
print(results.sort_values(by='RMSE'))
```

13.4 Code Output



```
--- Performance Metrics ---
              RMSE    MAPE (%)
Model
Holt-Winters    15.810381    2.207422
ARIMA(2,1,2)    55.222839    8.220491
Holt's Linear   84.384096   11.619686
SES             102.976535   14.251339
```

13.5 Interpretation & Conclusion

- PERFORMANCE ANALYSIS:**
 - SES** usually has the highest error because it fails to capture the trend and seasonality.
 - Holt's** captures the trend but misses the seasonal waves, resulting in moderate error.
 - Holt-Winters** typically performs best (lowest RMSE) for this dataset because it explicitly models the repeating seasonal cycles and the upward trend.
 - ARIMA** performs well but requires careful tuning; a basic (1,1,1) might not beat Holt-Winters on strongly seasonal data without seasonal differencing (SARIMA).
- Conclusion:** The **Holt-Winters** method is the best fit for this specific dataset due to its lowest RMSE score.